

JEFFERSON CITY, MO, USA

WMO: 724458

Lat: 38.591N

Lon: 92.156W

Elev: 175

StdP: 99.24

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03963

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.4	-11.3	-20.1	0.6	-12.8	-17.7	0.8	-10.1	11.0	2.5	9.7	0.3	2.9	290	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.2	24.5	33.7	24.1	32.4	23.8	26.3	32.1	25.6	31.3	24.9	30.5	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.2	29.4	23.9	19.1	28.7	23.0	18.1	27.9	82.7	32.3	79.4	31.6	76.4	30.4	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.2	7.3	DB	-18.7	37.5	3.0	1.8	-20.8	38.8	-22.5	39.9	-24.2	40.9	-26.3	42.2	(4)
(5)			WB	-19.1	27.6	2.9	1.3	-21.1	28.5	-22.8	29.2	-24.5	29.9	-26.5	30.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	0.2	2.4	7.9	14.0	19.0	23.9	26.0	25.0	20.9	14.4	8.0	2.1	(6)
(7)		DBStd	10.31	6.37	6.37	6.11	5.14	4.57	3.42	3.08	3.08	3.97	5.17	5.47	5.88	(7)
(8)		HDD10.0	1037	308	224	113	19	1	0	0	0	0	17	101	253	(8)
(9)		HDD18.3	2512	562	447	326	147	49	3	0	1	19	143	313	503	(9)
(10)		CDD10.0	2391	5	11	48	141	280	418	495	464	327	154	40	8	(10)
(11)		CDD18.3	826	0	0	3	17	70	172	237	207	96	22	2	0	(11)
(12)		CDH23.3	8326	0	2	36	194	638	1708	2599	2047	887	201	15	0	(12)
(13)		CDH26.7	3409	0	0	3	41	201	699	1202	893	319	50	1	0	(13)
(14)	Wind	WSAvg	3.0	3.4	3.5	3.7	3.9	3.2	2.9	2.3	2.1	2.2	2.7	3.1	3.2	(14)
(15)	Precipitation	PrecAvg	1011	45	48	76	98	127	109	89	87	102	84	80	61	(15)
(16)		PrecMax	1680	184	133	268	313	260	285	340	260	326	228	250	204	(16)
(17)		PrecMin	671	1	1	20	18	49	14	6	2	11	5	14	9	(17)
(18)		PrecStd	239	38	34	46	50	58	72	62	65	78	48	56	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	22.6	26.8	30.3	32.6	35.4	38.0	37.3	34.0	30.8	25.4	20.7	(19)
MCWB			13.8	14.9	17.2	19.0	21.9	24.1	23.9	24.4	23.4	21.8	16.7	16.4	(20)	
2%		DB	16.3	18.9	24.1	27.7	30.8	33.4	35.7	35.1	32.3	27.9	22.3	17.2	(21)	
		MCWB	11.0	12.7	15.7	17.6	21.7	23.9	24.8	24.3	23.2	19.4	15.1	12.9	(22)	
5%		DB	13.0	16.3	21.2	25.8	29.0	32.3	33.8	33.0	30.4	25.6	19.7	14.0	(23)	
		MCWB	8.5	10.5	14.2	16.8	21.0	23.5	24.6	23.9	22.4	18.2	14.1	9.7	(24)	
10%		DB	9.8	12.6	18.3	23.4	27.3	30.9	32.5	31.4	28.1	23.0	17.2	11.3	(25)	
		MCWB	6.2	8.0	12.7	15.9	20.0	23.0	24.4	23.6	21.3	17.4	12.4	7.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	16.2	18.8	20.6	24.1	26.3	27.4	27.4	25.2	23.1	18.5	16.7	(27)
MCDWB			18.8	20.5	24.6	26.8	29.8	31.9	32.1	32.1	31.4	28.2	23.0	20.3	(28)	
2%		WB	11.9	14.1	17.0	19.4	23.1	25.3	26.5	26.2	24.2	21.5	16.6	13.8	(29)	
		MCDWB	14.5	17.9	22.0	25.1	29.1	31.2	32.3	31.9	29.9	25.8	20.6	16.6	(30)	
5%		WB	8.8	11.2	15.4	18.2	21.9	24.4	25.8	25.2	23.4	20.0	14.9	10.7	(31)	
		MCDWB	12.4	15.3	20.1	23.7	27.1	30.1	31.5	30.8	28.3	23.6	18.8	13.2	(32)	
10%		WB	6.1	8.0	13.4	16.9	20.8	23.7	25.2	24.4	22.5	18.4	12.9	8.0	(33)	
		MCDWB	9.7	12.4	17.5	22.1	25.6	29.2	30.8	29.8	26.9	22.5	16.8	10.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.6	11.5	12.0	11.1	10.3	10.4	10.8	11.9	12.0	11.3	9.7	(35)
MCDBR			15.5	16.1	15.4	15.3	12.8	11.7	11.9	12.8	13.2	14.4	15.4	14.1	(36)	
MCWBR			10.8	10.6	8.5	7.8	5.9	4.6	4.1	4.7	5.5	7.2	9.5	10.0	(37)	
5% WB		MCDWB	13.6	14.8	12.7	11.9	11.1	10.2	10.4	10.8	11.0	11.2	12.8	12.6	(38)	
		MCWBR	10.4	10.7	8.1	7.1	5.7	4.7	4.6	4.7	5.1	6.5	9.2	10.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.317	0.344	0.393	0.434	0.436	0.457	0.462	0.394	0.350	0.310	0.298	(40)	
taud		2.463	2.414	2.401	2.273	2.220	2.253	2.199	2.195	2.354	2.468	2.523	2.498	(41)		
Ebn at Noon		878	909	915	884	851	847	824	810	851	861	864	858	(42)		
Edh at Noon		81	96	108	131	141	137	143	140	113	91	76	73	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.14	2.83	3.78	4.97	5.66	6.43	6.32	5.59	4.82	3.42	2.35	1.83	(44)	
(45)		RadStd	0.25	0.30	0.36	0.32	0.50	0.53	0.42	0.32	0.32	0.37	0.24	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-1.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82		

Nomenclature: See separate page